

श्वस रोग (Śvāsa Roga)

Introduction

There are innumerable number of diseases classifiable into two categories i : e *Sāmānya* & *Asāmānya* produced by the aggravation of three *Doṣās* as a result of three etiological factors. These are deadly diseases like *Sannipātika Jvara*, *Śaṅkhaka*, *Rohiṇi* etc. which can kill the patient. But none of these is as deadly as *Śvāsa* & *Hikkā* that can kill a patient instantaneously. Even if he has been suffering from various ailments, ultimately he falls a victim of *Śvāsa* & *Hikkā* that are immensely painful.

निरुक्ति एवं परिभाषा (Nirukti and Paribhāṣā)

Svāsa term indicates difficulty in breathing or shortness of breath. It is a symptom of many conditions that affect the *Prāṇvaha* srotas (respiratory system).

श्वसित्यनेनेति श्वासः ॥ *Śabdakalpdruma*

श्वसितौति श्वासः ॥ *Śabdakalpdruma*

कासः कुत्सित शब्दः ॥ *Vācaspatyam*

That which produces 'sigh' is called *Śvāsa*

श्वास इति वेगवदूर्ध्ववातत्रं ॥ *Ādḍ.Śa.Prā. 7/245*

The disease which originates from the forceful upward movement of *Vāta* is called *Śvāsa*

विहाय प्रकृतिं वायुः प्राणोऽथ कफसंयुतः ।

श्वासयत्यूर्ध्वगो भूत्वा तं श्वासं परिचक्षते ॥

Su.Utt. 51/4

प्राणो वायुः प्रकृतिं विहाय विगुणो भूत्वेत्यर्थः, ऊर्ध्वगो भूत्वा तथा कफसंयुतः सन् यदा श्वासयति तं बुधाः श्वासं परिचक्षते ॥

Su.Utt. 51/4 (Dalhaṇa)

The aggravated *Prāṇa Vāta* combines with *Kapha* moves *Ūrdhva Gati* (upwards) and produces gasping and laboured breathing which is called *Svāsa*.

श्वस रोग निदान (Nidāna of Śvāsa Roga)

पृथक् पञ्चविधावेतौ निर्दिष्टौ रोगसंग्रहे ।
तयोः शृणु समुत्थानं लिङ्गं च सभिषग्जितम् ॥
रजसा धूमवाताभ्यां शीतस्थानाम्बुसेवनात् ।
व्यायामादग्राम्यधर्माध्वरुक्षान्नविषमाशनात् ॥
आमप्रदोषादानाहाद्रोक्ष्यादत्यपतर्पणात् ।
दौर्बल्यान्मर्मणो घातादद्वन्द्वच्छुद्ध्यतियोगतः ॥
अतीसारज्वरच्छर्दिप्रतिश्यायक्षतक्षयात् ।
रक्तपित्तादुदावर्ताद्विसूच्यलसकादपि ॥
पाण्डुरोगाद्विषाच्चैव प्रवर्तते गदाविमौ ।
निष्पावमाषपिण्याकतिलतैलनिषेवणात् ॥
पिष्टशालूकविष्टम्भिभिदाहिगुरुभोजनात् ।
जलजानूपपिशितदध्यामक्षीरसेवनात् ॥
अभिष्यन्द्युपचाराच्च श्लेष्मलानां च सेवनात् ।
कण्ठोरसः प्रतीघाताद्विबन्धैश्च पृथग्विधैः ॥

C.S.Ci. 17/10-16

विदाहिगुरुविष्टम्भिरूक्षाभिष्यन्दिभोजनैः ।
शीतपानासनस्थानरजोधूमानिलानलैः ॥
व्यायामकर्मभाराध्ववेगाघातापतर्पणैः ।
आमदोषाभिघातस्त्रीक्ष्यदोषप्रपीडनैः ॥
विषमाशनाध्यशनैस्तथा समशनैरपि ।
हिक्का श्वासश्च कासश्च नृणां समुपजायते ॥

S.S.Utt. 50/3-5

The following etiological factors manifest Śvāsa and Hikkā Roga :- Exposure to dust, smoke, fire and wind; residing in a cold place & use of cold water; physical exertion; sexual intercourse & long walk beyond one's capacity; intake of rough food; irregular meals; manifestation of certain reactions inside the body due to underprocessed *Anna Rasa* (*Āma Pradoṣa*); distension of abdomen; roughness; fasting in excess; debility; injury to vital organs; usage of mutually contradictory substances; excessive administration of biopurificatory therapies.

Regular consumption of *Niṣpāva*, black gram, oilcake, sesame oilconsumption of flour preparations, tubers, constipating & burning substances, heavy food; intake of meat of aquatic

& marshy animals & birds; intake of curd & unboiled milk; consumption of channel blocking substances; regular intake of *Kapha* aggravating factors; injury to throat & chest and various other sorts of retention and suppression of natural urges.

These diseases manifest as complications in following diseases: *Atisāra*, *Jvara*, *Chardi*, *Pratiśyāya*, *Kṣata*, *Kṣaya*, *Raktapitta*, *Udāvarta*, *Visūcikā*, *Alasaka*, *Pāṇḍu* & *Viṣa*.

श्वास रोग संप्राप्ति (Samprāpti of Śvāsa Roga)

विहाय प्रकृति वायुः प्राणोऽथ कफसंयुतः ।
श्वासयत्पूर्वध्वगो भूत्वा तं श्वासं परिचक्षते ॥

S.S.Utt. 51/4

मारुतः प्राणवाहीनि स्रोतांस्याविश्य कुप्यति ।
उरःस्थः कफमुद्धूय हिक्काश्वासान् करोति सः ॥
घोरान् प्राणोपरोधाय प्राणिनां पञ्च पञ्च च ।

C.S.Ci. 17/17-18

यदा स्रोतांसि संरुध्य मारुतः कफपूर्वकः ।
विष्वग्ब्रजति संरुद्धस्तदा श्वासान्करोति सः ॥

C.S.Ci. 17/45

Due to consumption of etiological factors *Prāṇavāta* combining with *Kapha* leaves its own path and begins to move in the upward direction, giving rise to more of respirations is known as *Śvāsa Roga* & *Hikkā Roga*, which are of five kinds each & which may lead to the death of the patient.

कफवातात्मकावेतौ पित्तस्थानसमुद्भवौ ।
हृदयस्य रसादीनां धातूनां चोपशोषणौ ॥
तस्मात् साधारणावेतौ मतौ परमदुर्जयौ ।
मिथ्योपचरितौ क्रुद्धौ हत आशीविषाविव ॥

C.S.Ci. 17/8-9

Both *Hikkā* & *Śvāsa* originates from the seat of *Pitta* and are caused by the simultaneous aggravation of *Kapha* & *Vāta* and dries up the *Hṛdaya* & *Rasādi Dhātus*. Hence both being similar are regarded as very difficult to treat and if treated inappropriately leading to further aggravation of the condition & kill the patient like serpents.

Nidāna Sevana (Consumption of
Etiological factors)



Aggravation of *Kaphadi doṣas*



Development of *Āma* inside the gastro intestinal tract and *Śvāsa* originates from the seat of *Pitta*



Localisation of *Kapha doṣa* (especially *Rasadhātugata Malarupi Kapha*) in the *Prāṇavaha Srotas*



Leading to aggravation of *Vāta doṣa* inside the *Prāṇavaha Srotas* leaves its own path and begins to move in the upward direction



Dries up the *Hṛdaya* & *Rasādi Dhātus*.



Śvāsa Roga

श्वास रोग संप्राप्ति घटक (Samprāpti Ghataka of Śvāsa Roga)

Doṣa : *Vāta-Kapha* (*Prāṇa* and *Udāna* *Vāta*, *Bhodaka* & *Kledaka Kapha*)

Dūṣya : *Dhātu* → *Rasa Dhātu*
Upadhātu →
Śārīrika mala → *Mūtra*, *Purīṣa*
Dhātumala → *Kapha*, *Pitta*, *Tvacā*
and *Sveda*

Agni : *Jaṭharāgni*, *Dhātvāgni*

Agniduṣṭi : *Mandāgni*

Āma : *Āgnijanya*

Srotas : *Prāṇavaha*, *Udakavaha*, *Annavaha*
and *Rasavaha*

Srotoduṣṭi : *Saṅga* & *Vimārgamana*

Udbhavasthāna : *Āmāśayottha* (*Pittasthāna*)

Vyaktasthāna : *Āśya*, *Grīvā*, *Prṣṭa*, *Pārśva*

Sancārasthāna : *Prāṇavaha Sroto* *Avayava*

Adhiṣṭhāna : *Pitta Sthāna*

Prabhāva : *Kṣudra Śvāsa* → *Sādhyā*
Tamaka Śvāsa → *Kṛcchra*
Sādhyā Mahā, *Cinna* and
Urdvaśvāsa → *Asādhyā*

Svabhāva : *Āśukāri*

Rogamārga : *Madhyama Roga Mārga*

Roga Bheda : 5 kinds

पूर्वरूप (Pūrvārūpa)

प्राग्रूपं तस्य हृत्पीडा भक्तद्वेषोऽरतिः परा ।
आनाहः पार्श्वयोः शूलं वैरस्यं वदनस्य च ॥

S.S.Utt. 51/6

आनाहः पार्श्वशूलं च पीडनं हृदयस्य च ।
प्राणस्य च विलोमत्वं श्वासानां पूर्वलक्षणम् ॥

C.S.Ci. 17/20

Premonitory signs & symptoms of Śvāsa are as follows:- Pain in chest; aversion towards food; restlessness; flatulence; pain in the flanks; bad taste in the mouth and reversion of respiratory functions.

रोग भेद (Roga Bheda)

क्षुद्रकस्तमकश्छिन्नो महानूर्ध्वश्च पञ्चधा ।
भिद्यते स महाव्याधिः श्वास एको विशेषतः ॥

S.S.Utt. 51/5

Śvāsa is difficult to tolerate, incurable & dreadful disease and it is only one variety. But it is classified further into five kinds for easy understanding of the disease namely *Kṣudra*, *Tamaka*, *Chinna*, *Mahā* & *Ūrdhva*.

महाश्वास (Mahāśvāsa)

उद्धूयमानवातो यः शब्दवददुःखितो नरः ।
उच्चैः श्वसिति संरुद्धो मत्तर्षभ इवानिशम् ॥
प्रनष्टज्ञानविज्ञानस्तथा विभ्रान्तलोचनः ।
विकृताक्ष्याननो बद्धमूत्रवर्चा विशीर्णवाक् ॥
दीनः प्रश्वसितं चास्य दूराद्विज्ञायते भृशम् ।
महाश्वासोपसृष्टः स क्षिप्रमेव विपद्यते ॥
इति महाश्वासः ।

C.S.Ci. 17/46-48

निःसंज्ञः पार्श्वशूलार्तः शुष्ककण्ठोऽतिघोषवान् ।
संरुद्धनेत्रस्त्रायम्य यः श्वस्यात् स महान् स्मृतः ॥

S.S.Utt. 51/12

Mahāśvāsa is characterised by deep breath associated with discomfort and sound like that of mad bull or intoxicated bull due to obstruction in the *Prāṇavaha Srotas*. Because of this, *Vāta* moves forcibly in upward direction. Patient loses all sorts of knowledge, rolling eyes, distortion of eyes & face, retention of urine & stool, feeble voice, anxious expression, respiration sound becomes audible even from a distance. It is known as

Mahāśvāsa & succumbs to death instantaneously. Patient may become unconscious, pain in the flanks, dryness of throat, swollen eyes & stretching of body during breathing.

ऊर्ध्वश्वास (Ūrdhvaśvāsa)

दीर्घं श्वसिति यस्तूर्ध्वं न च प्रत्याहरत्यधः ।
श्लेष्मावृतमुखस्रोताः कुण्डलगन्धवहार्दितः ॥
ऊर्ध्वदृष्टिर्विपश्यंश्च विभ्रान्ताक्ष इतस्ततः ।
प्रमुहान् वेदनार्तश्च शुष्कास्योऽरतिपीडितः ॥
ऊर्ध्वश्वासे प्रकुपिते ह्यधःश्वासो निरुध्यते ।
मुह्यतस्ताम्यतश्चोर्ध्वं श्वासस्तस्यैव हन्त्यसून् ॥
इत्यूर्ध्वश्वासः । C.S.Ci. 17/49-51

मर्मस्वायम्यमानेषु श्वसन्मूढो मुहुश्च यः ।
ऊर्ध्वप्रेक्षी हतरवस्तमूर्ध्वश्वासमादिशेत् ॥

S.S.Utt. 51/13

Ūrdhvaśvāsa is characterised by forceful expiration and impeded inspiration because of which the patient becomes unconscious with tremors and finally dies. It is associated with prolonged expiration & inability to have inspiration, adhesion of sputum in mouth & *Prāṇavaha Srotas*, affliction with aggravated *Vāta*, gazes abnormally with eyes up & rolling, unconsciousness, suffering from severe pain, dryness of mouth, restlessness, patient feels stretching kind of pain in vital parts (*Marmās*), faints oftenly after respiration and loss of voice.

छिन्नश्वास (Chinnaśvāsa)

यस्तु श्वसिति विच्छिन्नं सर्वप्राणेन पिडितः ।
न वा श्वसिति दुःखार्तो मर्मच्छेदरुगर्दितः ॥
आनाहस्वेदमूर्च्छार्तो दह्यमानेन बस्तिना ।
विप्लुताक्षः परिक्षीणः श्वसन् रक्तैकलोचनः ॥
विचेताः परिशुष्कास्यो विवर्णः प्रलपन्नरः ।
छिन्नश्वासेन विच्छिन्नः स शीघ्रं प्रजाहात्यसून् ॥
इति छिन्नश्वासः । C.S.Ci. 18/52-54

आध्मातो दह्यमानेन बस्तिना सरुजं नरः ।
सर्वप्राणेन विच्छिन्नं श्वस्याच्छिन्नं तमादिशेत् ॥

S.S.Utt. 51/11

Chinnaśvāsa is characterised by intermittent respiration or no respiration at all or painful

respiration due to affliction of all the channels carrying *Prāṇavāta*; cutting pain in vital parts; *Ānāha*; perspiration; fainting; burning sensation in urinary bladder; tearing eyes; extreme emaciation; redness in one eye; detraction of mind; dryness in the mouth; abnormal complexion; delirium. Patient succumbs to death instantaneously due to cut off by the intermittent respiration.

तमकश्वास (Tamaka Śvāsa)

प्रतिलोमं यदा वायुः स्रोतांसि प्रतिपद्यते ।
ग्रीवां शिरश्च संगृह्य श्लेष्माणं समुदीर्य च ॥
करोति पीनसं तेन रुद्धो घुर्घुरकं तथा ।
अतीव तीव्रवेगं च श्वासं प्राणप्रपीडकम् ॥
प्रताम्यत्यतिवेगाच्च कासते सन्निरुध्यते ।
प्रमोहं कासमानश्च स गच्छति मुहुर्मुहुः ॥
श्लेष्मण्यमुच्यमाने तु भृशं भवति दुःखितः ।
तस्यैव च विमोक्षान्ते मुहूर्तं लभते सुखम् ॥
अथास्योद्ध्वंसते कण्ठः कृच्छ्राच्छक्नोति भाषितुम् ।
न चापि निद्रां लभते शयानः श्वासपीडितः ॥
पार्श्वे तस्यावर्गीति शयानस्य समीरणः ।
आसीनो लभते सौख्यमुष्णं चैवाभिनन्दति ॥
उच्छ्रिताक्षो ललाटेन स्विद्यता भृशमर्तिमान् ।
विशुष्कास्यो मुहुः श्वासो मुहुश्चैवावधम्यते ॥
मेघाम्बुशीतप्राग्वातैः श्लेष्मलैश्चाभिवर्धते ।
स याप्यस्तमकश्वासः साध्यो वा स्यान्नवोत्थितः ॥
इति तमकश्वासः । C.S.Ci. 17/55-62

तृट्स्वेदवमथुप्रायः कण्ठघुर्घुरिकान्वितः ।
विशेषाद् दुर्दिने ताम्येच्छवासः स तमको मतः ॥
घोषेण महताऽऽविष्टः सकासः सकफो नरः ।
यः श्वसित्यबलोऽन्नद्विद् सुप्तस्तमकपीडितः ॥

S.S.Utt. 51/8-9

Due to obstruction in *Prāṇavaha Srotas*, *Vāta* moves in reverse order and pervades the entire *Prāṇavaha Srotas* and stimulates the *Kapha* in the throat & head causing rhinitis. Further it manifest wheezing sound; greater enhancement of respiration in relation to rate & rhythm; feeling of darkness in front of the eyes; thirst; vomiting; paroxysmal attack of this disease are severe enough to put the patient in distress, to bend him forward, to cough again & again which sometimes result in fainting; until the expectoration of provoked *Kapha* patient remains

agony and patient finds temporary relief; after expectorating *Kapha* patient is unable to talk much because of choking in the throat; aversion to food; does not get sleep when he lies down because of catching of breath; finds comfort in warm substances; eyes wide opened; sweating in forehead; restlessness; dryness of the mouth; suffers frequently from paroxysms of dyspnoea which aggravates during cloudy days, by the use of cold water, breeze from east and consumption of *Kapha* aggravating factors. It is manageable in chronic stage and curable if it is of recent onset.

प्रतमक श्वास (Pratamaka Śvāsa) & संतमक श्वास (Santamaka Śvāsa)

स शाम्यति कफे हीने स्वपतश्च विवर्द्धते ।
मूर्च्छाज्वराभिभूतस्य ज्ञेयः प्रतमकस्तु सः ॥
S.S.Utt. 51/10

ज्वरमूर्च्छापरीतस्य विद्यात् प्रतमकं तु तम् ।
उदावर्तरजोऽजीर्णक्लिन्नकायनिरोधजः ॥
तमसा वर्धतेऽत्यर्थं शीतैश्चाशु प्रशाम्यति ।
मज्जस्तमसीवाऽस्य विद्यात् संतमकं तु तम् ॥
इति प्रतमकसंतमकश्वासौ । C.S.Ci. 10/63-64

Disease which subsides when *Kapha* decreases & increases greatly when sleeping and associated with fainting & fever is known as *Pratamaka*. It is caused by *Udāvarta*, dust, indigestion, humidity, suppression of natural urges by an old person. It manifest due to aggravation of *Pitta*.

Santamaka Śvāsa manifest due to increase of *Tamas* and gets relieved quickly by cooling regimes. These patients feels as if they submerged in darkness.

क्षुद्र श्वास (Kṣudra Śvāsa)

रुक्षायामोद्धवः कोष्ठे क्षुद्रो वात उदीरयन् ।
क्षुद्रश्वासो न सोऽत्यर्थं दुःखेनाङ्गप्रबाधकः ॥
हिनस्ति न स गात्राणि न च दुःखो यथेतरे ।
न च भोजनपानानां निरुणद्ध्युचितां गतिम् ॥
नेन्द्रियाणां व्यथां नापि कांचिदापादयेद्भुजम् ।
स साध्य उक्तो बलिनः सर्वे चाव्यक्तलक्षणाः ॥
C.S.Ci. 51/765-67

किञ्चिदारभमाणस्य यस्य श्वासः प्रवर्तते ।

निषण्णस्यैति शान्तिञ्च स क्षुद्र इति संज्ञितः ॥

S.S.Utt. 51/7

Kṣudra Śvāsa manifest due to mild aggravation of *Vāta* in the *Koṣṭha* on account of physical exertion and rough regimens. It subsides when person takes rest. It does not produce discomfort, does not afflict body parts, it is not troublesome as other forms of *Śvāsa* are, does not restrict the normal movement of foods & drinks, does not produce any discomfort or pain in sense organs. It is curable.

Prognosis

All other forms of *Śvāsa* are curable when they are mild & affecting strong persons.

साध्यासाध्यता (Sādhyāsādhyaṭā)

एषां प्राणहरा वर्ज्या घोरास्ते ह्याशुकारिणः ॥
भेषजैः साध्ययाप्यास्तु क्षिप्रं भिषगुपाचरेत् ।
उपेक्षिता दहेयुर्हि शुष्कं वृक्षमिवानलः ॥

C.S.Ci. 17/68-69

Incurable varieties of *Śvāsa* are fatal because they are very severe & galloping. That's why these kinds of *Śvāsa* should not be treated.

On the contrary curable & manageable varieties of *Śvāsa* should be treated with proper medicaments because if neglected or delayed they may reduce the body to ashes like the fire burning the dried tree.

क्षुद्रः साध्यतमस्तेषां तमकः कृच्छ्र उच्यते ।
त्रयः श्वासा न सिध्यन्ति तमको दुर्बलस्य च ॥

S.S.Utt. 51/14

Kṣudra Śvāsa is curable easily, *tamaka Śvāsa* is difficult to cure, *Ūrdhva*, *Maha*, *Chinna Śvāsa* in weak patients is incurable.

उपशय/पथ्य (Upaśaya/Pathya)

The useful items in the case of *dyspnoea* includes the following: Sweating, smoking herbal drugs, sleep during daytime, intake of such items as the red variety of rice, *Kulattha* (dolichos bean), wheat, barley, flesh of rabbit, peacock, partridge, wood-cutter, chicken, parrot and the like birds and

animals of the arid regions, old *Ghṛta*, milk and *Ghṛta* of the goat, liquor, honey, leaves of *Nidigdhikā*, *Vāstuka* (chenopodium), *Taṇḍulīya*, *Jivantī*, young plant of the radish, *Paṭola*, brinjal, garlic, *Haraḍa*, *Jambīra* lemon, *Bimbī* fruits, *Mātuluṅga* lemon, date fruits, smaller cardamom, *Puṣkaramūla*, hot water, *Trikaṭu*, urine of the cow and all other items that reduces the *Kapha* and *Vāta* contents of the body. Also helpful to the patient of *Śvāsa* is the cauterization on both sides of his breast, on the middle fingers of his hands as well as around the pit of his throat. (B.R. 16/132-136)

अनुपशय/अपथ्य (Anupaśaya/Apathya)

Harmful activities and items in cases of *Śvāsa roga* includes : Suppression of the natural urges, nasal therapies, the *Vasti* therapy, chewing wood sticks with the teeth, hard labour, excessive walking, lifting heavy items, facing dust, moving out in sun, consuming meals heavy to digest, having intercourse, taking hot items of meal, taking meat of the animals from the marshy region, taking deep fried nuggets and the like dishes, taking items that produce *Kapha*, taking the *Urada* beans, blood-letting, facing the easternly wind, drinking water immediately after eating meals, taking the milk and *Ghṛta* of sheep, drinking impure water, eating fishes, vegetable leaves of *Kanda* and the like tubers, leaves of mustard, etc. (B.R. 16/137-139).

MCQs

1. *Amapradosha anaha roukshyadanyapatr-panat* is the Nidana of
 - a. Shwasa
 - b. Gulma
 - c. Grahani
 - d. Pravahika
2. *Rakthaikalocana* is seen in
 - a. Chinna Shwasaa
 - b. Mahashwasa
 - c. Urdhwashwasa
 - d. Tamakashwasa
3. *Dosha* responsible for *Shwasa* is
 - a. Prana-Udana
 - b. Udana-Vyanaa
 - c. Prana-Kapha
 - d. Apana-Prana
4. *Viplutaksha parikshina* mentioned in context to
 - a. Chinna Shwasa
 - b. Mahashwasa
 - c. Urdhwashwasa
 - d. Tamakashwasa

5. *Pranastajnana vijnana statha vibhranta lochana* is the feature of
 a. *Chinna Shwasa* b. *Mahashwasa*
 c. *Urdhwashwasa* d. *Tamakashwasa*
6. *Mattarshabha ivanisham* is the feature of
 a. *Tamakashwasa* b. *Mahashwasa*
 c. *Urdhwashwasa* d. *Chinna Shwasa*
7. *Deenaha prashwasitam chasya duradvijn-ayate bhrusham* observed in
 a. *Chinna Shwasaa* b. *Urdhwashwasa*
 c. *Mahashwasa* d. *Tamakashwasa*
8. *Srotas* involved in the *Samprapti* of *Swasa*
 a. *Pranavaha* b. *Mutravaha*
 c. *Udaka vaha* d. *Anna vaha*
9. *Anaha sweda murcharto dahyamanena bastina* is the feature of
 a. *Chinna Shwasaa* b. *Mahashwasa*
 c. *Urdhwashwasa* d. *Mutrakricchra*
10. *Sa shamyati kaphe hine swapatascha vivardhate* feature observed in
 a. *Pratamaka swasha*
 b. *Santamaka swasha*
 c. *Khudra swasha*
 d. *Tamakashwasha*
11. *Tritswedavamathu prayahkanthaghur-ghurikanvitah* feature observed in
 a. *Pratamaka swasha*
 b. *Santamaka swasha*
 c. *Khudra swasha*
 d. *Tamakashwasha*

Answer

1.	a	2.	a	3.	c
4.	a	5.	a	6.	b
7.	c	8.	a	9.	a
10.	a	11.	d		